2

CLASSIFYING TOPOSES AND THE ‘BRIDGE’ TECHNIQUE

This chapter consists of two parts. In the first part we review the fundamental notion of classifying topos of a geometric theory and discuss the appropriate kinds of interpretations between theories which induce morphisms between the associated classifying toposes; the theoretical presentation is accompanied by a few concrete examples of classifying toposes of theories naturally arising in mathematics. We then establish a characterization theorem for universal models of geometric theories inside classifying toposes. In the second part we explain the general unifying technique ‘toposes as bridges’ originally introduced in [14]. This technique, which allows one to extract ‘concrete’ information from the existence of different representations for the classifying topos of a geometric theory, will be systematically exploited in the course of the book to establish theoretical results as well as applications.

2.1 **Geometric logic and classifying toposes**

2.1.1 *Geometric theories*

In this book we shall be mostly concerned with geometric theories, because of their connection with the theory of Grothendieck toposes.

Geometric theories are linked to Grothendieck toposes via the notion of classifying topos. Indeed, as we shall see in section 2.1.2, every geometric theory has a classifying topos and conversely any Grothendieck topos can be seen as the classifying topos of some geometric theory.

Recall from section 1.2 that a *geometric theory* over a first-order signature Σ is a theory whose axioms are sequents of the form $(\phi \vdash_{\vec{x}} \psi)$, where $\phi(\vec{x})$ and $\psi(\vec{x})$ are *geometric formulae* over Σ in the context $\vec{x}$ i.e. formulae with a finite number of free variables in $\vec{x}$ built up from atomic formulae over Σ by only using finitary conjunctions, infinitary disjunctions and existential quantifications.

Whilst the notion of geometric theory might seem quite restrictive at first sight, it turns out that most of the (first-order) theories naturally arising in mathematics have a geometric axiomatization (over their signature). Also, the possibility of employing infinitary disjunctions (of an arbitrary cardinality) in the construction of geometric formulae makes geometric logic particularly expressive and suitable for axiomatizing theories which do not belong to the realm of classical first-order logic (think for example of the notion of algebraic extension of a given field, or of that of ℓ-group with strong unit).

Theories, Sites, Toposes. Olivia Caramello.

Rather remarkably, if a finitary first-order theory $\mathbb{T}$ is not geometric, one can canonically construct a coherent theory $\mathbb{T}'$ over a larger signature, called the *Morleyization* of $\mathbb{T}$, whose models in the category of sets (and, more generally, in any Boolean coherent category) can be identified with those of $\mathbb{T}$:

Proposition 2.1.1 (Lemma D1.5.13 [55], cf. also [2]). *Let $\mathbb{T}$ be a first-order theory over a signature Σ. Then there is a signature Σ' containing Σ and a coherent theory $\mathbb{T}'$ over Σ', called the* Morleyization *of $\mathbb{T}$, such that we have*

$$\mathbb{T}\text{-mod}_e(\mathcal{C}) \simeq \mathbb{T}'\text{-mod}(\mathcal{C})$$

for any Boolean coherent category $\mathcal{C}$.

Sketch of proof The signature Σ' of $\mathbb{T}'$ has, in addition to all the sorts, function symbols and relation symbols of the signature Σ of $\mathbb{T}$, two relation symbols $C_\phi \rightarrowtail A_1 \cdots A_n$ and $D_\phi \rightarrowtail A_1 \cdots A_n$ for each first-order formula ϕ over Σ (where $A_1 \cdots A_n$ is the string of sorts corresponding to the canonical context of ϕ), while the axioms of $\mathbb{T}'$ are given by the sequents of the form $(C_\phi \vdash_{\vec{x}} C_\psi)$ for any axiom $(\phi \vdash_{\vec{x}} \psi)$ of $\mathbb{T}$, plus a set of coherent sequents involving the new relation symbols C_ϕ and D_ϕ which ensure that in any model M of $\mathbb{T}'$ in a Boolean coherent category $\mathcal{C}$, the interpretation of C_ϕ coincides with the interpretation of ϕ and the interpretation of D_ϕ coincides with the complement of the interpretation of ϕ (cf. pp. 859-860 of [55] for more details). □

Proposition 2.1.2 *Let $\mathbb{T}$ be a finitary first-order theory over a signature Σ and $\mathbb{T}'$ its Morleyization. Then*

(i) *For any finitary first-order sequent $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ over Σ, σ is provable in $\mathbb{T}$ using classical first-order logic if and only if the sequent $(C_\phi \vdash_{\vec{x}} C_\psi)$ is provable in $\mathbb{T}'$ using coherent logic.*

(ii) *The classical first-order syntactic category $\mathcal{C}_{\mathbb{T}}^{\text{fo-cl}}$ of $\mathbb{T}$ is isomorphic to the coherent syntactic category of $\mathbb{T}'$ and to the classical first-order syntactic category of $\mathbb{T}'$.*

Proof

(i) This follows immediately from the axioms defining $\mathbb{T}'$.

(ii) It is readily seen that every finitary first-order (resp. coherent) formula over the signature of $\mathbb{T}$ is classically provably equivalent (resp. provably equivalent in coherent logic) to a coherent formula over the signature of $\mathbb{T}'$. It follows that the coherent syntactic category of $\mathbb{T}'$ is Boolean and that every morphism between models of $\mathbb{T}'$ in Boolean coherent categories is an elementary morphism. Hence, by Proposition 2.1.1, both the coherent syntactic category of $\mathbb{T}'$ and the classical first-order syntactic category of $\mathbb{T}'$ satisfy the universal property of the category $\mathcal{C}_{\mathbb{T}}^{\text{fo-cl}}$ with respect to models of $\mathbb{T}$ in Boolean coherent categories whence, by universality, these categories are both equivalent to $\mathcal{C}_{\mathbb{T}}^{\text{fo-cl}}$ (cf. Remark 1.4.11), as required.

□

Remark 2.1.3 It follows at once from the proof of part (ii) of the proposition that a first-order theory is complete in the sense of classical model theory (i.e. every first-order sentence over its signature is either provably false or provably true, but not both) if and only if its Morleyization is complete in the sense of geometric logic (i.e. every geometric sentence over its signature is either provably false or provably true, but not both).

The notion of Morleyization is important because it enables us to study any kind of first-order theory by using the methods of topos theory. In fact, we can expect many important properties of first-order theories to be naturally expressible as properties of their Morleyizations, and these latter properties to be in turn expressible in terms of invariant properties of their classifying toposes (cf. for instance Remarks 2.1.3 and 4.2.22). Still, one can 'turn' a finitary first-order theory $\mathbb{T}$ into a geometric one in alternative ways, i.e. by simply adding some sorts to the signature of $\mathbb{T}$ and axioms over the extended signature so as to ensure that each of the first-order formulae which appear in the axioms of $\mathbb{T}$ becomes equivalent to a geometric formula in the new theory and the set-based models of the latter can be identified with those of $\mathbb{T}$. The Morleyization construction just represents a canonical, generally 'non-economical' way of doing this which works uniformly for any finitary first-order theory.

2.1.2 *The notion of classifying topos*

We have seen in section 1.3.1 (cf. Proposition 1.3.15) that every Grothendieck topos is a geometric category. Thus we can consider models of geometric theories in any Grothendieck topos. Inverse image functors of geometric morphisms of toposes preserve finite limits (by definition) and arbitrary colimits (having a right adjoint); so they are geometric functors and hence they preserve the interpretation of (arbitrary) geometric formulae (cf. the proof of Lemma 1.3.23). In general such functors are *not* Heyting functors, which explains why the following definition only makes sense for geometric theories.

Definition 2.1.4 Let $\mathbb{T}$ be a geometric theory over a given signature. A *classifying topos* of $\mathbb{T}$ is a Grothendieck topos $\mathbf{Set}[\mathbb{T}]$ (also denoted by $\mathcal{E}_{\mathbb{T}}$) such that for any Grothendieck topos $\mathcal{E}$ we have an equivalence of categories

$$\mathbf{Geom}(\mathcal{E}, \mathbf{Set}[\mathbb{T}]) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$$

natural in $\mathcal{E}$.

Naturality means that for any geometric morphism $f : \mathcal{E} \to \mathcal{F}$, we have a commutative square (up to natural isomorphism)

$$\begin{tikzcd}
\mathbf{Geom}(\mathcal{F}, \mathbf{Set}[\mathbb{T}]) \arrow[r, "\simeq"] \arrow[d, "-\circ f"] & \mathbb{T}\text{-mod}(\mathcal{F}) \arrow[d, "\mathbb{T}\text{-mod}(f^*)"] \\
\mathbf{Geom}(\mathcal{E}, \mathbf{Set}[\mathbb{T}]) \arrow[r, "\simeq"] & \mathbb{T}\text{-mod}(\mathcal{E})
\end{tikzcd}$$

in the (meta-)2-category $\mathfrak{CAT}$ of categories.

In other words, there is a model U of $\mathbb{T}$ in $\mathcal{E}_{\mathbb{T}}$, called 'the' *universal model* of $\mathbb{T}$, characterized by the universal property that any model M in a Grothendieck topos can be obtained, up to isomorphism, as the pullback $f^*(U)$ of U along the inverse image f^* of a unique (up to isomorphism) geometric morphism f from $\mathcal{E}$ to $\mathbf{Set}[\mathbb{T}]$.

Remark 2.1.5 The classifying topos of a geometric theory $\mathbb{T}$ can be seen as a *representing object* for the (pseudo-)functor

$$\mathbb{T}\text{-mod} : \mathfrak{BTop}^{\text{op}} \to \mathfrak{CAT}$$

which assigns

- to a topos $\mathcal{E}$ the category $\mathbb{T}\text{-mod}(\mathcal{E})$ of models of $\mathbb{T}$ in $\mathcal{E}$ and
- to a geometric morphism $f : \mathcal{E} \to \mathcal{F}$ the functor $\mathbb{T}\text{-mod}(f^*) : \mathbb{T}\text{-mod}(\mathcal{F}) \to \mathbb{T}\text{-mod}(\mathcal{E})$ sending a model $M \in \mathbb{T}\text{-mod}(\mathcal{F})$ to its image $f^*(M)$ under f^*.

In particular, classifying toposes are unique up to categorical equivalence.

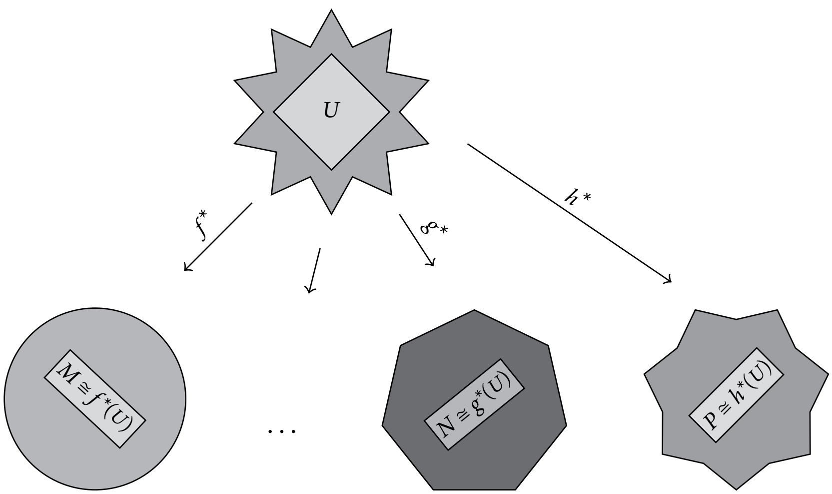


Fig. 2.1: Classifying topos

In Figure 2.1, the big shapes represent different toposes while the inner lighter shapes represent models of a given theory inside them; in particular, the dark grey star represents the classifying topos of a given theory and the light grey diamond represents 'the' universal model of the theory inside it. As the picture illustrates, all the models of the theory, including all the classical set-based models, are sorts of 'shadows' of the universal model lying in the classifying topos. This indicates that the symmetries of the theory are best understood by adopting the point of view of its classifying topos, since all the other models are images of this inner 'core' under 'deformations' realized by structure-preserving functors.

Contrary to what happens in classical logic, where one is forced to appeal to non-constructive principles such as the axiom of choice in order to ensure the existence of 'enough' set-based models for faithfully representing a finitary first-order theory, here we dispose of a constructively defined universe in which the syntax and semantics of a geometric theory meet yielding a strong form of completeness (cf. Theorem 1.4.5) and definability (cf. Theorem 6.1.3). Moreover, classifying toposes not only exist for finitary geometric theories, but also for arbitrary infinitary ones which do not necessarily satisfy a classical completeness theorem (one can exhibit non-contradictory infinitary geometric theories without any set-based models).

Definition 2.1.6 A *universal model* of a geometric theory $\mathbb{T}$ is a model $U_{\mathbb{T}}$ of $\mathbb{T}$ in a Grothendieck topos $\mathcal{G}$ such that for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{F}$ there exists a unique (up to isomorphism) geometric morphism $f_M : \mathcal{F} \to \mathcal{G}$ such that $f_M^*(U_{\mathbb{T}}) \cong M$.

Remarks 2.1.7 (a) By the (2-dimensional) Yoneda lemma, if a Grothendieck topos $\mathcal{G}$ contains a universal model of a geometric theory $\mathbb{T}$ then $\mathcal{G}$ satisfies the universal property of the classifying topos of $\mathbb{T}$. Conversely, if a Grothendieck topos $\mathcal{E}$ classifies a geometric theory $\mathbb{T}$ then $\mathcal{E}$ contains a universal model of $\mathbb{T}$.

(b) If M and N are universal models of a geometric theory $\mathbb{T}$ lying respectively in toposes $\mathcal{F}$ and $\mathcal{G}$ then there exists a unique up to isomorphism (geometric) equivalence between $\mathcal{F}$ and $\mathcal{G}$ such that its inverse image functors send M and N to each other (up to isomorphism).

The classifying topos of a geometric theory can be canonically built as the category of sheaves on its geometric syntactic category with respect to the geometric topology on it. For smaller fragments of geometric logic, such as for example cartesian (resp. regular, coherent) logic, there exist variants of this syntactic construction, consisting in replacing the geometric syntactic site of the theory with its cartesian (resp. regular, coherent) syntactic site. More specifically, the following result holds (cf. also section D3.1 of [55]).

Theorem 2.1.8

(i) *For any universal Horn theory* $\mathbb{T}$*, the topos* $[(\mathcal{C}_{\mathbb{T}}^{\mathrm{alg}})^{\mathrm{op}}, \mathbf{Set}]$ *classifies* $\mathbb{T}$.
(ii) *For any cartesian theory* $\mathbb{T}$*, the topos* $[(\mathcal{C}_{\mathbb{T}}^{\mathrm{cart}})^{\mathrm{op}}, \mathbf{Set}]$ *classifies* $\mathbb{T}$.
(iii) *For any regular theory* $\mathbb{T}$*, the topos* $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}}, J_{\mathbb{T}}^{\mathrm{reg}})$ *classifies* $\mathbb{T}$.
(iv) *For any coherent theory* $\mathbb{T}$*, the topos* $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ *classifies* $\mathbb{T}$.
(v) *For any geometric theory* $\mathbb{T}$*, the topos* $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ *classifies* $\mathbb{T}$.

Proof The proof of (i) requires a slightly different argument from that of the other points and can be found in [7] (cf. Theorem 1 therein).

By Diaconescu's equivalence (cf. Theorem 1.1.34) and Proposition 1.1.32, for any cartesian (resp. regular, coherent, geometric) theory $\mathbb{T}$ the geometric morphisms $\mathcal{E} \to [(\mathcal{C}_{\mathbb{T}}^{\mathrm{cart}})^{\mathrm{op}}, \mathbf{Set}]$ (resp. $\mathcal{E} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}}, J_{\mathbb{T}}^{\mathrm{reg}})$, $\mathcal{E} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$,

$\mathcal{E} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$) correspond, naturally in $\mathcal{E}$, to the cartesian functors $\mathcal{C}_{\mathbb{T}}^{\text{cart}} \to \mathcal{E}$ (resp. to the cartesian $J_{\mathbb{T}}^{\text{reg}}$-continuous functors $\mathcal{C}_{\mathbb{T}}^{\text{reg}} \to \mathcal{E}$, to the cartesian $J_{\mathbb{T}}^{\text{coh}}$-continuous functors $\mathcal{C}_{\mathbb{T}}^{\text{coh}} \to \mathcal{E}$, to the cartesian $J_{\mathbb{T}}$-continuous functors $\mathcal{C}_{\mathbb{T}} \to \mathcal{E}$). So (ii), (iii), (iv) and (v) follow from Theorem 1.4.10 and Proposition 1.4.8. □

Remark 2.1.9 For each of the points of Theorem 2.1.8, the image of the universal model $M_{\mathbb{T}}$ of $\mathbb{T}$ in its syntactic category (as in Definition 1.4.4) under the Yoneda embedding from it to the classifying topos of $\mathbb{T}$ yields a universal model $U_{\mathbb{T}}$ of $\mathbb{T}$ (recall from Remark 1.4.9 that all the ‘logical’ topologies are subcanonical). It follows in particular from Theorem 1.4.5(ii) that $U_{\mathbb{T}}$ is a conservative model of $\mathbb{T}$.

As a corollary one obtains the following fundamental result due to A. Joyal, G. Reyes and M. Makkai:

Theorem 2.1.10 *Every geometric theory has a classifying topos.* □

In the converse direction, we have the following well-known result:

Theorem 2.1.11 *Every Grothendieck topos is the classifying topos of some geometric theory.*

Proof Let $\mathbf{Sh}(\mathcal{C}, J)$ be a Grothendieck topos, where $(\mathcal{C}, J)$ is a small site. Diaconescu’s equivalence provides, for any Grothendieck topos $\mathcal{E}$, an equivalence, natural in $\mathcal{E}$, between the category $\mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}, J))$ of geometric morphisms from $\mathcal{E}$ to $\mathbf{Sh}(\mathcal{C}, J)$ and the category $\mathbf{Flat}_J(\mathcal{C}, \mathcal{E})$ of J-continuous flat functors from $\mathcal{C}$ to $\mathcal{E}$. Now, we can construct a geometric theory $\mathbb{T}_J^{\mathcal{C}}$ such that its models in any Grothendieck topos $\mathcal{E}$ can be identified precisely with the J-continuous flat functors from $\mathcal{C}$ to $\mathcal{E}$ (and the homomorphisms of $\mathbb{T}_J^{\mathcal{C}}$-models with the natural transformations between the corresponding flat functors); clearly, $\mathbb{T}_J^{\mathcal{C}}$ will be classified by the topos $\mathbf{Sh}(\mathcal{C}, J)$. We shall call $\mathbb{T}_J^{\mathcal{C}}$ the *theory of J-continuous flat functors on $\mathcal{C}$*. It is instructive to write down an axiomatization for $\mathbb{T}_J^{\mathcal{C}}$.

The signature of $\mathbb{T}_J^{\mathcal{C}}$ has one sort $\ulcorner a \urcorner$ for each object a of $\mathcal{C}$ and one function symbol $\ulcorner f \urcorner : \ulcorner a \urcorner \to \ulcorner b \urcorner$ for each arrow $f : a \to b$ in $\mathcal{C}$. The axioms of $\mathbb{T}_J^{\mathcal{C}}$ are the following (to indicate that a variable x has sort $\ulcorner a \urcorner$ we write x^a):

$$(\top \vdash_x \ulcorner f \urcorner(x) = x)$$

for any identity arrow f in $\mathcal{C}$;

$$(\top \vdash_x \ulcorner f \urcorner(x) = \ulcorner h \urcorner(\ulcorner g \urcorner(x)))$$

for any triple of arrows f, g, h of $\mathcal{C}$ such that f is equal to the composite $h \circ g$;

$$\left(\top \vdash_{[]} \bigvee_{a \in Ob(\mathcal{C})} (\exists x^a)\top\right)$$

(where the disjunction ranges over all the objects of $\mathcal{C}$);

$$\left(\top \vdash_{x^a, y^b} \bigvee_{a \xleftarrow{f} c \xrightarrow{g} b} (\exists z^c)(\ulcorner f \urcorner(z^c) = x^a \wedge \ulcorner g \urcorner(z^c) = y^b)\right)$$

for any objects a, b of $\mathcal{C}$ (where the disjunction ranges over all the cones $a \xleftarrow{f} c \xrightarrow{g} b$ on the discrete diagram on the two objects a and b);

$$\left(\ulcorner f \urcorner(x^a) = \ulcorner g \urcorner(x^a) \vdash_{x^a} \bigvee_{\substack{c \xrightarrow{h} a \,\mid \\ f \circ h = g \circ h}} (\exists z^c)(\ulcorner h \urcorner(z^c) = x^a)\right)$$

for any pair of arrows $f, g : a \to b$ in $\mathcal{C}$ with common domain and codomain (where the disjunction ranges over all the arrows h which equalize f and g);

$$\left(\top \vdash_{x^a} \bigvee_{i \in I} (\exists y_i^{b_i})(\ulcorner f_i \urcorner(y_i^{b_i}) = x^a)\right)$$

for each J-covering family $\{f_i : b_i \to a \mid i \in I\}$.

Notice that the first two groups of axioms express functoriality, the third, fourth and fifth the filtering property (cf. Proposition 1.1.32), and the sixth the property of J-continuity. □

Two geometric theories are said to be *Morita-equivalent* if they have equivalent classifying toposes (cf. Definition 2.2.1 below). From the above discussion it follows that Grothendieck toposes can be thought of as canonical representatives for Morita equivalence classes of geometric theories.

It is important to note that the method of constructing classifying toposes via syntactic sites is by no means the only one for 'calculating' the classifying topos of a geometric theory. Alternative techniques, of more 'semantic' or 'geometric' nature, have been developed. For instance, as we shall see in Chapter 3, every representation of a geometric theory as an extension of another geometric theory over the same signature leads to a representation of its classifying topos as a subtopos of the classifying topos of the latter theory; when applied to extensions $\mathbb{S}$ of theories $\mathbb{T}$ classified by a presheaf topos, this leads to a 'semantic' representation for the classifying topos of $\mathbb{S}$ as a topos of sheaves on the opposite of the category of finitely presentable $\mathbb{T}$-models (cf. Chapter 8). More generally, as we shall argue in section 2.2, it is reasonable to expect 'different ways of looking at a certain theory' to materialize into different representations of its classifying topos.

It should also be mentioned that the notions of geometric theory and classifying topos can be 'relativized' to an arbitrary base topos (cf. for instance sections 6.5 of [50] and B4.2 of [55]). Still, a theory of internal geometric theories and relative classifying toposes has not yet been formally developed.

Via the syntactic construction of classifying toposes of coherent theories, the classical completeness theorem for coherent theories (cf. Theorem 1.4.18) translates into the following theorem about coherent toposes. Recall that a Grothendieck topos is said to be *coherent* if it admits a small site of definition of the form $(\mathcal{C}, J)$, where $\mathcal{C}$ is a cartesian category and J is a finite-type topology on $\mathcal{C}$ (cf. section 8.1.4 for the notion of topology of finite type).

Theorem 2.1.12 (Deligne's theorem – Exposé VI, Proposition 9.0 [3]). *Assuming the axiom of choice, every coherent topos has enough points.* □

2.1.3 *Interpretations and geometric morphisms*

The assignment from a collection of geometric theories to a collection of Grothendieck toposes sending a theory to its classifying topos can be made functorial, as follows. A natural notion of morphism between theories is given by the notion of interpretation of one theory into another. As we shall see below, there are actually many variants of this notion, all of which induce geometric morphisms between the respective classifying toposes.

Definition 2.1.13 Let $\mathbb{T}_1$ and $\mathbb{T}_2$ be two cartesian (resp. regular, coherent, geometric) theories.

(a) A *cartesian* (resp. *regular*, *coherent*, *geometric*) *interpretation* of $\mathbb{T}_1$ in $\mathbb{T}_2$ is a cartesian (resp. regular, coherent, geometric) functor from $\mathcal{C}^{\text{cart}}_{\mathbb{T}_1}$ (resp. $\mathcal{C}^{\text{reg}}_{\mathbb{T}_1}$, $\mathcal{C}^{\text{coh}}_{\mathbb{T}_1}$, $\mathcal{C}_{\mathbb{T}_1}$) to $\mathcal{C}^{\text{cart}}_{\mathbb{T}_2}$ (resp. $\mathcal{C}^{\text{reg}}_{\mathbb{T}_2}$, $\mathcal{C}^{\text{coh}}_{\mathbb{T}_2}$, $\mathcal{C}_{\mathbb{T}_2}$).

(b) The cartesian (resp. regular, coherent, geometric) theories $\mathbb{T}_1$ and $\mathbb{T}_2$ are said to be *cartesianly bi-interpretable* (resp. *regularly bi-interpretable*, *coherently bi-interpretable*, *geometrically bi-interpretable*) if their cartesian (resp. regular, coherent, geometric) syntactic categories are equivalent.

If $\mathbb{T}_1$ and $\mathbb{T}_2$ are cartesian theories then any cartesian functor $\mathcal{C}^{\text{cart}}_{\mathbb{T}_1} \to \mathcal{C}^{\text{cart}}_{\mathbb{T}_2}$ induces a geometric morphism

$$[(\mathcal{C}^{\text{cart}}_{\mathbb{T}_2})^{\text{op}}, \mathbf{Set}] \to [(\mathcal{C}^{\text{cart}}_{\mathbb{T}_1})^{\text{op}}, \mathbf{Set}]$$

(cf. Example 1.1.15(c)).

If $\mathbb{T}_1$ and $\mathbb{T}_2$ are regular (resp. coherent, geometric) theories then, since by Proposition 1.4.8 the regular (resp. coherent, geometric) functors $\mathcal{C}^{\text{reg}}_{\mathbb{T}_1} \to \mathcal{C}^{\text{reg}}_{\mathbb{T}_2}$ (resp. $\mathcal{C}^{\text{coh}}_{\mathbb{T}_1} \to \mathcal{C}^{\text{coh}}_{\mathbb{T}_2}$, $\mathcal{C}_{\mathbb{T}_1} \to \mathcal{C}_{\mathbb{T}_2}$) are exactly the cartesian functors which send $J^{\text{reg}}_{\mathbb{T}_1}$-covering sieves (resp. $J^{\text{coh}}_{\mathbb{T}_1}$-covering sieves, $J_{\mathbb{T}_1}$-covering sieves) on $\mathcal{C}^{\text{reg}}_{\mathbb{T}_1}$ (resp. $\mathcal{C}^{\text{coh}}_{\mathbb{T}_1}$, $\mathcal{C}_{\mathbb{T}_1}$) to $J^{\text{reg}}_{\mathbb{T}_2}$-covering sieves (resp. $J^{\text{coh}}_{\mathbb{T}_2}$-covering sieves, $J_{\mathbb{T}_2}$-covering sieves) on $\mathcal{C}^{\text{reg}}_{\mathbb{T}_2}$ (resp. $\mathcal{C}^{\text{coh}}_{\mathbb{T}_2}$, $\mathcal{C}_{\mathbb{T}_2}$), any regular (resp. coherent, geometric) interpretation of $\mathbb{T}_1$ in $\mathbb{T}_2$ induces a geometric morphism $\mathbf{Sh}(\mathcal{C}^{\text{reg}}_{\mathbb{T}_2}, J^{\text{reg}}_{\mathbb{T}_2}) \to \mathbf{Sh}(\mathcal{C}^{\text{reg}}_{\mathbb{T}_1}, J^{\text{reg}}_{\mathbb{T}_1})$ (resp. $\mathbf{Sh}(\mathcal{C}^{\text{coh}}_{\mathbb{T}_2}, J^{\text{coh}}_{\mathbb{T}_2}) \to \mathbf{Sh}(\mathcal{C}^{\text{coh}}_{\mathbb{T}_1}, J^{\text{coh}}_{\mathbb{T}_1})$, $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}_2}, J_{\mathbb{T}_2}) \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}_1}, J_{\mathbb{T}_1})$) (again by Example 1.1.15(c)).

Let us now discuss the relationship between syntactic categories and classifying toposes. We shall see that, whilst it is always possible to recover the cartesian syntactic category of a cartesian theory from its classifying topos (up to equivalence), it is only possible in general to recover the effectivization of the regular syntactic category (resp. the effective positivization of the coherent syntactic category) of a regular (resp. coherent) theory from its classifying topos.

The topos-theoretic invariants which are useful in this respect are the following:

Definition 2.1.14 (a) An object A of a Grothendieck topos is said to be *irreducible* if every epimorphic family in the topos with codomain A contains a split epimorphism.

(b) An object A of a Grothendieck topos is said to be *compact* if every epimorphic family with codomain A contains a finite epimorphic subfamily.

(c) An object A of a Grothendieck topos is said to be *supercompact* if every epimorphic family with codomain A contains an epimorphism.

(d) An object A of a Grothendieck topos $\mathcal{E}$ is said to be *coherent* if it is compact and for any arrow $u : B \to A$ in $\mathcal{E}$ such that B is compact, the domain of the kernel pair of u is compact.

(e) An object A of a Grothendieck topos $\mathcal{E}$ is said to be *supercoherent* if it is supercompact and for any arrow $u : B \to A$ in $\mathcal{E}$ such that B is supercompact, the domain of the kernel pair of u is supercompact.

For a proof of the following proposition we refer the reader to section D3.3 of [55] for points (ii) and (iii) and to section 4 of [20] for point (i); for the notions involved in the proposition, see section 1.3.2.

Proposition 2.1.15

(i) *The cartesian syntactic category* $\mathcal{C}_{\mathbb{T}}^{\mathrm{cart}}$ *of a cartesian theory* $\mathbb{T}$ *can be recovered, up to equivalence, from its classifying topos* $[(\mathcal{C}_{\mathbb{T}}^{\mathrm{cart}})^{\mathrm{op}}, \mathbf{Set}]$ *as the full subcategory on the irreducible objects.*

(ii) *The effectivization* $\mathbf{Eff}(\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}})$ *of the regular syntactic category* $\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}}$ *of a regular theory* $\mathbb{T}$ *can be recovered, up to equivalence, from its classifying topos* $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}}, J_{\mathbb{T}}^{\mathrm{reg}})$ *as the full subcategory on the supercoherent objects.*

(iii) *The effective positivization* $\mathbf{Eff}(\mathbf{Pos}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}))$ *of the coherent syntactic category* $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}$ *of a coherent theory* $\mathbb{T}$ *can be recovered, up to equivalence, from its classifying topos* $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ *as the full subcategory on the coherent objects.*

Notice that every object of $\mathbf{Eff}(\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}})$ has a syntactic description as a 'formal quotient' of an object of $\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}}$ by an equivalence relation in $\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}}$, and similarly any object of $\mathbf{Eff}(\mathbf{Pos}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}))$ is a 'formal quotient' in $\mathbf{Pos}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}})$ of a 'formal finite coproduct' of objects of $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}$ by an equivalence relation in $\mathbf{Pos}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}})$.

This motivates the following definition:

Definition 2.1.16 (a) A *generalized regular interpretation* of a regular theory $\mathbb{T}_1$ in a regular theory $\mathbb{T}_2$ is a regular functor $\mathbf{Eff}(\mathcal{C}_{\mathbb{T}_1}^{\mathrm{reg}}) \to \mathbf{Eff}(\mathcal{C}_{\mathbb{T}_2}^{\mathrm{reg}})$.

(b) A *generalized coherent interpretation* of a coherent theory $\mathbb{T}_1$ in a coherent theory $\mathbb{T}_2$ is a coherent functor $\mathbf{Eff}(\mathbf{Pos}(\mathcal{C}_{\mathbb{T}_1}^{\mathrm{coh}})) \to \mathbf{Eff}(\mathbf{Pos}(\mathcal{C}_{\mathbb{T}_2}^{\mathrm{coh}}))$.

As is the case for their classical counterparts, generalized interpretations also induce geometric morphisms between the classifying toposes of the two theories.

For geometric theories, as we saw in section 1.3.2, the analogue of the effective positivization construction is the ∞-pretopos completion: applied to the syntactic category of a geometric theory, this construction yields precisely the classifying topos of the theory (by Giraud's theorem and Theorem 1.4.10).

The following proposition represents the 'functorialization' of Proposition 2.1.15. Before stating it, we need to introduce some definitions.

A geometric morphism $f : \mathcal{E} \to \mathcal{F}$ is said to be *coherent* if the inverse image functor $f^* : \mathcal{F} \to \mathcal{E}$ of f sends coherent objects of $\mathcal{F}$ to coherent objects of $\mathcal{E}$. Similarly, a geometric morphism $f : \mathcal{E} \to \mathcal{F}$ is said to be *regular* if $f^* : \mathcal{F} \to \mathcal{E}$ sends supercoherent objects of $\mathcal{F}$ to supercoherent objects of $\mathcal{E}$.

Proposition 2.1.17

(i) *Every essential geometric morphism $\mathcal{E}_{\mathbb{T}_2} \to \mathcal{E}_{\mathbb{T}_1}$ between the classifying toposes of two cartesian theories $\mathbb{T}_1$ and $\mathbb{T}_2$ is induced by a (unique up to isomorphism) cartesian interpretation of $\mathbb{T}_1$ in $\mathbb{T}_2$.*

(ii) *Every regular geometric morphism $\mathcal{E}_{\mathbb{T}_2} \to \mathcal{E}_{\mathbb{T}_1}$ between the classifying toposes of two regular theories $\mathbb{T}_1$ and $\mathbb{T}_2$ is induced by a (unique up to isomorphism) generalized regular interpretation of $\mathbb{T}_1$ in $\mathbb{T}_2$.*

(iii) *Every coherent geometric morphism $\mathcal{E}_{\mathbb{T}_2} \to \mathcal{E}_{\mathbb{T}_1}$ between the classifying toposes of two coherent theories $\mathbb{T}_1$ and $\mathbb{T}_2$ is induced by a (unique up to isomorphism) generalized coherent interpretation of $\mathbb{T}_1$ in $\mathbb{T}_2$.*

Note that, since the classifying topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ of a geometric theory $\mathbb{T}$ is the ∞-pretopos completion of $\mathcal{C}_{\mathbb{T}}$, we can regard any geometric morphism $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}_2}, J_{\mathbb{T}_2}) \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}_1}, J_{\mathbb{T}_1})$ between the classifying toposes of two geometric theories $\mathbb{T}_1$ and $\mathbb{T}_2$ as a generalized geometric interpretation of $\mathbb{T}_1$ in $\mathbb{T}_2$.

We can give an alternative description of the notions of interpretation defined above in terms of internal models, as follows. If $\mathbb{T}$ is a cartesian (resp. regular, coherent, geometric) theory then, by Theorem 1.4.10, the category of cartesian (resp. regular, coherent, geometric) functors from $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$ (resp. $\mathcal{C}_{\mathbb{T}}^{\text{reg}}$, $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$, $\mathcal{C}_{\mathbb{T}}$) to any cartesian (resp. regular, coherent, geometric) category $\mathcal{D}$ is equivalent to the category of models of $\mathbb{T}$ in $\mathcal{D}$, where the equivalence sends each model $M \in \mathbb{T}\text{-mod}(\mathcal{D})$ to the functor $F_M : \mathcal{C}_{\mathbb{T}} \to \mathcal{D}$ assigning to each object $\{\vec{x} \,.\, \phi\}$ of $\mathcal{C}_{\mathbb{T}}$ its interpretation $[[\vec{x} \,.\, \phi]]_M$ in M. Thus, any geometric interpretation of a geometric theory $\mathbb{T}_1$ in a geometric theory $\mathbb{T}_2$ corresponds to an internal $\mathbb{T}_1$-model in the syntactic category $\mathcal{C}_{\mathbb{T}_2}$ of $\mathbb{T}_2$, and similarly for cartesian, regular and coherent theories (and their generalized interpretations). Note that if $\mathbb{T}_2$ has enough set-based models (for example, if $\mathbb{T}_2$ is coherent assuming the axiom of choice) then the condition that a certain Σ_1-structure U, where Σ_1 is the signature of $\mathbb{T}_1$, in (a completion of) the syntactic category of $\mathbb{T}_2$ be a $\mathbb{T}_1$-model is equivalent to the requirement that for any $\mathbb{T}_2$-model M in $\mathbf{Set}$ the functor F_M send U to a $\mathbb{T}_1$-model.

Remarks 2.1.18 (a) Since a given theory may belong to more than one fragment of geometric logic, it is natural to wonder whether the notions of bi-interpretability of theories given above for different fragments of geometric logic are compatible with each other. We shall answer this question in the affirmative in section 2.2.4.

(b) Since the classification of models of geometric theories already takes place at the level of geometric categories (cf. Theorem 1.4.10), one may naturally wonder why we should focus our attention on more sophisticated structures such as classifying toposes. In fact, there are several reasons for this:
First, a Grothendieck topos is, in terms of internal categorical structure, much richer than an arbitrary geometric category and hence constitutes a mathematical environment which is naturally more amenable to computations and which enjoys higher degrees of symmetry. In fact, it is a standard process in mathematics to complete mathematical entities with respect to natural operations through the formal addition of 'imaginaries' and work in the extended, more 'symmetric' environment to solve problems posed in the original context (think for instance of the real line and the complex plane).
Second, a very important aspect of Grothendieck toposes, on which the theory of topos-theoretic 'bridges' described in section 2.2 is based, is the fact that they admit a very well-behaved, although highly non-trivial, representation theory. The abstract relationship between an object and its different 'presentations' is geometrically embodied in the context of Grothendieck toposes in the form of relationships between a given topos and its different sites of definition (or, more generally, its different representations); the possibility of working at two levels, that of toposes and that of their presentations, makes the theory extremely fruitful and technically flexible.
Third, the 2-category $\mathfrak{Btop}$ of Grothendieck toposes is itself quite rich in terms of (2-)categorical structure (cf. for instance section B4 of [55]).

2.1.4 *Classifying toposes for propositional theories*

Recall that a *propositional theory* is a geometric theory over a signature which has no sorts. The signature of a propositional theory thus merely consists of a set of 0-ary relation symbols, which define the atomic propositions; all the other formulae (i.e. sentences) are built up from them by using finitary conjunctions and infinitary disjunctions.

Propositional theories can be used for axiomatizing subsets of a given set satisfying particular properties, e.g. filters on a frame or prime ideals of a commutative ring.

Proposition 2.1.19 (cf. Remark D3.1.14 [55]). *Localic toposes are precisely the classifying toposes of propositional theories.*

Specifically, we can define, for any locale L, a propositional theory $\mathbb{P}_L$ classified by it, as follows. The signature of $\mathbb{P}_L$ consists of one atomic proposition F_a (to be thought of as the assertion that a is in the filter) for each $a \in L$; the axioms of $\mathbb{P}_L$ are

$$(\top \vdash F_1),$$

all the sequents of the form

$$(F_a \wedge F_b \vdash F_{a\wedge b})$$

for any $a, b \in L$, and all the sequents of the form

$$\left(F_a \vdash \bigvee_{i\in I} F_{a_i}\right)$$

whenever $\bigvee_{i\in I} a_i = a$ in L.

Notice that the set-based models of the theory $\mathbb{P}_L$ are the completely prime filters on L, that is the subsets $F \subseteq L$ such that $1 \in F, a \wedge b \in F$ whenever $a, b \in F$ and for any family of elements $\{a_i \in L \mid i \in I\}$ such that $\bigvee_{i\in I} a_i \in F$, there exists $i \in I$ such that $a_i \in F$.

Remark 2.1.20 Recall that the *Lindenbaum-Tarski algebra* $\mathcal{L}_{\mathbb{T}}$ of a propositional theory $\mathbb{T}$ is the algebra consisting of the provable-equivalence classes of sentences over its signature. In the case of a geometric propositional theory $\mathbb{T}$, $\mathcal{L}_{\mathbb{T}}$ is clearly a frame, equivalent to the geometric syntactic category of $\mathbb{T}$. The classifying topos of $\mathbb{T}$ is given by the category $\mathbf{Sh}(\mathcal{L}_{\mathbb{T}}, J_{\mathcal{L}_{\mathbb{T}}})$ of sheaves on this locale with respect to the canonical topology $J_{\mathcal{L}_{\mathbb{T}}}$ on it. This indicates that the notion of classifying topos of a geometric theory represents a natural first-order generalization of that of Lindenbaum-Tarski of a propositional theory.

We shall encounter propositional theories again in section 2.1.6.

2.1.5 *Classifying toposes for cartesian theories*

We can describe the classifying topos of a cartesian theory directly in terms of its finitely presented models.

Recall that, given a cartesian theory $\mathbb{T}$ over a signature Σ, a $\mathbb{T}$-model M in **Set** is said to be *finitely presented* by a cartesian formula-in-context $\phi(\vec{x})$ over Σ if there exists a string of elements $(\xi_1, \dots, \xi_n) \in MA_1 \times \cdots \times MA_n$ (where $A_1 \cdots A_n$ is the string of sorts associated with $\vec{x}$), called the *generators* of M, such that for any $\mathbb{T}$-model N in **Set** and any string of elements $(b_1, \dots, b_n) \in NA_1 \times \cdots \times NA_n$ such that $(b_1, \dots, b_n) \in [[\vec{x} \,.\, \phi]]_N$, there exists a unique arrow $f : M \to N$ in $\mathbb{T}$-mod(**Set**) such that $(f_{A_1} \times \cdots \times f_{A_n})((\xi_1, \dots, \xi_n)) = (b_1, \dots, b_n)$.

We denote by f.p.$\mathbb{T}$-mod(**Set**) the full subcategory of $\mathbb{T}$-mod(**Set**) on the finitely presented models.

Theorem 2.1.21 (Corollary D3.1.2 [55]). *For any cartesian theory $\mathbb{T}$, we have an equivalence of categories*

$$\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}) \simeq (\mathcal{C}_{\mathbb{T}}^{\text{cart}})^{\text{op}}.$$

In particular, $\mathbb{T}$ is classified by the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ *(cf. Theorem 2.1.8(ii)).*

Remark 2.1.22 If $\mathbb{T}$ is a universal Horn theory (in particular, a finitary algebraic theory) then the category $\mathcal{C}_{\mathbb{T}}^{\mathrm{alg}}$ is dual to the category of finitely presented $\mathbb{T}$-algebras, and the presheaf topos $[(\mathcal{C}_{\mathbb{T}}^{\mathrm{alg}})^{\mathrm{op}}, \mathbf{Set}]$ satisfies the universal property of the classifying topos for $\mathbb{T}$ (cf. Theorem 1 of [7]).

Examples 2.1.23

(a) The (algebraic) theory of Boolean algebras is classified by the topos $[\mathbf{Bool}_{\mathrm{fin}}, \mathbf{Set}]$ where $\mathbf{Bool}_{\mathrm{fin}}$ is the category of finite Boolean algebras and Boolean algebra homomorphisms between them.

(b) The (algebraic) theory of commutative rings with unit is classified by the topos $[\mathbf{Rng}_{\mathrm{f.g.}}, \mathbf{Set}]$, where $\mathbf{Rng}_{\mathrm{f.g.}}$ is the category of finitely generated rings and ring homomorphisms between them.

2.1.6 *Further examples*

2.1.6.1 *The Zariski topos*

Let Σ be the one-sorted signature for the theory $\mathbb{T}$ of commutative rings with unit, i.e. the signature consisting of two binary function symbols $+$ and $\cdot$, one unary function symbol $-$ and two constants 0 and 1.

The *coherent theory of local rings* is obtained from $\mathbb{T}$ by adding the sequents

$$(0 = 1 \vdash_{[]} \bot)$$

and

$$((\exists z)((x + y) \cdot z = 1) \vdash_{x,y} ((\exists z)(x \cdot z = 1) \vee (\exists z)(y \cdot z = 1))).$$

Definition 2.1.24 The *Zariski topos* is the topos $\mathbf{Sh}(\mathbf{Rng}_{\mathrm{f.g.}}^{\mathrm{op}}, J)$ of sheaves on the opposite of the category $\mathbf{Rng}_{\mathrm{f.g.}}$ of finitely generated rings with respect to the topology J on $\mathbf{Rng}_{\mathrm{f.g.}}^{\mathrm{op}}$ defined as follows: for any cosieve S in $\mathbf{Rng}_{\mathrm{f.g.}}$ on an object A, $S \in J(A)$ if and only if S contains a finite family $\{\xi_i : A \to A[{s_i}^{-1}] \mid 1 \leq i \leq n\}$ of canonical inclusions $\xi_i : A \to A[{s_i}^{-1}]$ in $\mathbf{Rng}_{\mathrm{f.g.}}$ where $\{s_1, \ldots, s_n\}$ is a set of elements of A which is not contained in any proper ideal of A.

Proposition 2.1.25 (Theorem VII.6.3 [63]). *The (coherent) theory of local rings is classified by the Zariski topos.*

More generally, one can prove that the big Zariski topos of an affine scheme Spec(A) classifies the theory of local A-algebras. On the other hand, the small Zariski topos Spec(A) classifies the theory of localizations of the ring A, equivalently the propositional theory $\mathbb{P}_A$ of *prime filters* on A defined as follows. The signature of $\mathbb{P}_A$ consists of a propositional symbol P_a for each element $a \in A$, and the axioms of $\mathbb{P}_A$ are the following:

$$(\top \vdash P_{1_A});$$
$$(P_{0_A} \vdash \bot);$$
$$(P_{a \cdot b} \dashv\vdash P_a \wedge P_b)$$

for any a, b in A;

$$(P_{a+b} \vdash P_a \vee P_b)$$

for any $a, b \in A$.

The models of $\mathbb{P}_A$ in **Set** are precisely the prime filters on A, that is the subsets S of A such that the complement $A \setminus S$ is a prime ideal. In fact, the Zariski topology is homeomorphic to the subterminal topology (in the sense of Definition 1.1.23) on the set of points of the classifying topos of $\mathbb{P}_A$. If, instead of taking the theory $\mathbb{P}_A$ of prime filters, we had considered the propositional theory of prime ideals (axiomatized over the same signature in the obvious way), we would have obtained a classifying topos inequivalent to the small Zariski topos of A, in spite of the fact that the two theories have the same models in **Set**. In fact, in order to prove that their categories of set-based models are equivalent, one has to use the Law of Excluded Middle, a principle which is not sound in general toposes, even in very simple ones such as the Sierpiński topos (i.e. the topos of sheaves on the Sierpiński space).

2.1.6.2 *The classifying topos for integral domains*

The *theory of integral domains* is the theory obtained from the theory of commutative rings with unit by adding the axioms

$$(0 = 1 \vdash_{[]} \bot);$$

$$(x \cdot y = 0 \vdash_{x,y} x = 0 \vee y = 0).$$

Proposition 2.1.26 *The theory of integral domains is classified by the topos $\mathbf{Sh}(\mathbf{Rng}_{\text{f.g.}}^{\text{op}}, J)$ of sheaves on the opposite of the category $\mathbf{Rng}_{\text{f.g.}}$ of finitely generated rings with respect to the topology J on $\mathbf{Rng}_{\text{f.g.}}^{\text{op}}$ defined as follows: for any cosieve S in $\mathbf{Rng}_{\text{f.g.}}$ on an object A, $S \in J(A)$ if and only if either*

- *A is the zero ring and S is the empty sieve on it, or*
- *S contains a non-empty finite family $\{\pi_{a_i} : A \to A/(a_i) \mid 1 \leq i \leq n\}$ of canonical projections $\pi_{a_i} : A \to A/(a_i)$ in $\mathbf{Rng}_{\text{f.g.}}$ where $\{a_1, \ldots, a_n\}$ is a set of elements of A such that $a_1 \cdot \ldots \cdot a_n = 0$.*

For a proof of this proposition, see section 8.1.5 below.

2.1.6.3 *The topos of simplicial sets*

Definition 2.1.27 The theory $\mathbb{I}$ of *intervals* is written over a one-sorted signature having a binary relation symbol $\leq$ and two constants b and t, and has as axioms the following sequents:

$$(\top \vdash_x x \leq x);$$
$$(x \leq y \wedge y \leq z \vdash_{x,y,z} x \leq z);$$
$$(x \leq y \wedge y \leq x \vdash_{x,y} x = y);$$
$$(\top \vdash_x x \leq t \wedge b \leq x);$$
$$(b = t \vdash \bot);$$
$$(\top \vdash_{x,y} x \leq y \vee y \leq x).$$

Proposition 2.1.28 *The theory $\mathbb{I}$ is classified by the topos $[\Delta^{\mathrm{op}}, \mathbf{Set}]$ of simplicial sets.*

A detailed proof of this proposition may be found in section VIII.8 of [63].

2.1.7 *A characterization theorem for universal models in classifying toposes*

It is natural to wonder whether, given a geometric theory $\mathbb{T}$, one can give conditions on a pair $(\mathcal{E}, M)$ consisting of a Grothendieck topos $\mathcal{E}$ and a model M of $\mathbb{T}$ in $\mathcal{E}$ for $\mathcal{E}$ to be a classifying topos of $\mathbb{T}$ and M a universal $\mathbb{T}$-model in it. The following result provides a positive answer to this question.

Theorem 2.1.29 *Let $\mathbb{T}$ be a geometric theory, $\mathcal{E}$ a Grothendieck topos and M a model of $\mathbb{T}$ in $\mathcal{E}$. Then $\mathcal{E}$ is a classifying topos of $\mathbb{T}$ and M a universal model of $\mathbb{T}$ if and only if the following conditions are satisfied:*

(i) *The family $\mathcal{F}$ of objects which can be built from the interpretations in M of the sorts, function symbols and relation symbols over the signature of $\mathbb{T}$ by using geometric logic constructions (i.e. the objects given by the domains of the interpretations in M of geometric formulae over the signature of $\mathbb{T}$) is separating for $\mathcal{E}$.*

(ii) *The model M is conservative for $\mathbb{T}$, that is for any geometric sequent σ over the signature of $\mathbb{T}$, σ is valid in M if and only if it is provable in $\mathbb{T}$.*

(iii) *Any arrow k in $\mathcal{E}$ between objects A and B in the family $\mathcal{F}$ of point (i) is definable; that is, if A (resp. B) is equal to the interpretation of a geometric formula $\phi(\vec{x})$ (resp. $\psi(\vec{y})$) over the signature of $\mathbb{T}$, there exists a $\mathbb{T}$-provably functional formula θ from $\phi(\vec{x})$ to $\psi(\vec{y})$ such that the interpretation of θ in M is equal to the graph of k.*

Proof The fact that conditions (i), (ii) and (iii) are necessary for $\mathcal{E}$ to be a classifying topos of $\mathbb{T}$ and M a universal $\mathbb{T}$-model in it follows at once from Theorem 2.1.8 and Remark 2.1.9; so it remains to prove that they are also sufficient.

By the universal property of the geometric syntactic category $\mathcal{C}_{\mathbb{T}}$ of $\mathbb{T}$, the $\mathbb{T}$-model M corresponds to a geometric functor $F_M : \mathcal{C}_{\mathbb{T}} \to \mathcal{E}$ assigning to each object of $\mathcal{C}_{\mathbb{T}}$ (the domain of) its interpretation in M and acting accordingly on arrows. Condition (ii) in the statement of the theorem is equivalent to the assertion that the functor F_M be faithful, while condition (iii) is equivalent to saying that F_M is full. Therefore under conditions (ii) and (iii) $\mathcal{C}_{\mathbb{T}}$ embeds, up to equivalence, as a full subcategory of $\mathcal{E}$. Now, condition (i) ensures that $\mathcal{C}_{\mathbb{T}}$ is dense in $\mathcal{E}$, whence the Comparison Lemma (Theorem 1.1.8) yields an equivalence $\mathcal{E} \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J)$, where J is the Grothendieck topology on $\mathcal{C}_{\mathbb{T}}$ induced by the canonical topology on $\mathcal{E}$, that is the geometric topology on $\mathcal{C}_{\mathbb{T}}$. By the syntactic construction of classifying toposes and universal models, we can thus conclude that $\mathcal{E}$ is 'the' classifying topos of $\mathbb{T}$ and M is a universal model for $\mathbb{T}$. □

As immediate corollaries of Theorem 2.1.29, we recover the following known results:

- Let $\mathcal{C}$ be a separating set of objects for a Grothendieck topos $\mathcal{E}$, and $\Sigma^{\mathrm{f}}_{\mathcal{C}}$ the signature consisting of one sort $\ulcorner c \urcorner$ for each object c of $\mathcal{C}$ and one function symbol $\ulcorner f \urcorner$ for each arrow f in $\mathcal{E}$ between objects in $\mathcal{C}$ (cf. section 1.3.1.1). Then there exists a geometric theory $\mathbb{T}$ over the signature $\Sigma^{\mathrm{f}}_{\mathcal{C}}$ classified by $\mathcal{E}$, whose universal model is given by the 'tautological' $\Sigma^{\mathrm{f}}_{\mathcal{C}}$-structure in $\mathcal{E}$ (cf. p. 837 of [55]).
- Let B be a pre-bound for $\mathcal{E}$ over **Set** (that is, an object such that the subobjects of its finite powers form a separating set of objects for $\mathcal{E}$); then there exists a one-sorted geometric theory $\mathbb{T}$ classified by $\mathcal{E}$ and a universal model of $\mathbb{T}$ in it whose underlying object is B (cf. Theorem D3.2.5 [55]).

The first result can be obtained from Theorem 2.1.29 by observing that the theory $\mathbb{T}$ of the tautological $\Sigma^{\mathrm{f}}_{\mathcal{C}}$-structure satisfies all the hypotheses of the theorem; that the first two conditions are satisfied is obvious, while the fact that the third holds can be proved as follows. Since every object of the syntactic category of $\mathbb{T}$ is a subobject of a finite product of objects of the form $\{x^{\ulcorner c \urcorner} . \top\}$ (for $c \in \mathcal{C}$), it is enough to prove that every arrow $k : R \to c$ in $\mathcal{E}$ having as domain the interpretation $R = [[\vec{x} . \phi]]_M$ of a geometric formula $\phi(\vec{x})$ over the signature of $\mathbb{T}$ and as codomain an object c in $\mathcal{C}$ is definable. Suppose that the sorts of the variables in $\vec{x} = (x_1, \ldots, x_n)$ are respectively $\ulcorner c'_1 \urcorner, \ldots, \ulcorner c'_n \urcorner$ and let $r : R \to c'_1 \times \cdots \times c'_n$ denote the corresponding subobject. Since $\mathcal{C}$ is a separating set of objects for $\mathcal{E}$, the family of arrows $\{ f_i : c_i \to R \mid i \in I\}$ from objects of $\mathcal{C}$ to R is epimorphic, whence the geometric formula

$$\bigvee_{i \in I} (\exists z_i^{\ulcorner c_i \urcorner})(x_1 = \ulcorner g_1 \urcorner(z_i) \wedge \cdots \wedge x_n = \ulcorner g_n \urcorner(z_i) \wedge x^{\ulcorner c \urcorner} = \ulcorner k \circ f_i \urcorner(z_i)),$$

where $r \circ f_i = < g_1, \ldots, g_n >$, is $\mathbb{T}$-provably functional from $\{\vec{x} . \phi\}$ to $\{x^{\ulcorner c \urcorner} . \top\}$ and its interpretation coincides with k. By Diaconescu's equivalence, the theory $\mathbb{T}$ can be explicitly characterized as the theory of flat $J^{\mathcal{C}}_{\mathcal{E}}$-continuous functors on $\mathcal{C}$, where $J^{\mathcal{C}}_{\mathcal{E}}$ is the Grothendieck topology on $\mathcal{C}$ induced by the canonical topology on $\mathcal{E}$.

The second result can be deduced from Theorem 2.1.29 by taking $\mathbb{T}$ to be the theory of the tautological structure over the one-sorted signature Σ_B consisting of one n-ary relation symbol $\ulcorner R \urcorner$ for each subobject $R \rightarrowtail B^n$ in $\mathcal{E}$. The fact that $\mathbb{T}$ satisfies the first two conditions of the theorem is obvious, while the validity of the third condition follows from the fact that the graphs of morphisms $R \rightarrowtail B^n \to B$ in $\mathcal{E}$ are interpretations of $(n+1)$-ary relation symbols over Σ_B.

2.2 Toposes as 'bridges'

In this section we describe the unifying methodology 'toposes as bridges' introduced in [16]. This general technique, which has already found applications in different fields of mathematics (see [28]), will be systematically applied throughout the book, either implicitly or explicitly, in a variety of different contexts, so it is essential for the reader to acquire familiarity with it.

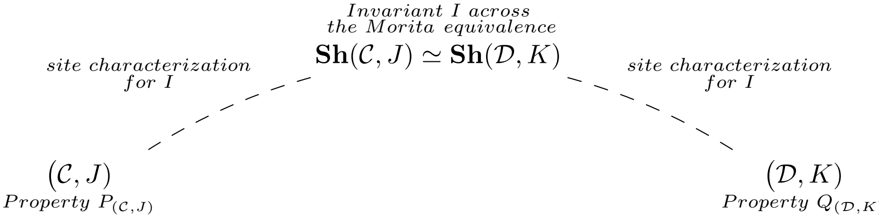
2.2.1 *The 'bridge-building' technique*

The 'bridge-building' technique allows one to construct topos-theoretic 'bridges' connecting distinct mathematical theories with each other.

Specifically, if $\mathbb{T}$ and $\mathbb{T}'$ are two Morita-equivalent theories (that is, geometric theories classified by the same topos) then their common classifying topos can be used as a 'bridge' for transferring information between them:

$$
\begin{tikzcd}
 & \mathcal{E}_{\mathbb{T}} \simeq \mathcal{E}_{\mathbb{T}'} \arrow[dr, dashed, bend left] & \\
\mathbb{T} \arrow[ur, dashed, bend left] & & \mathbb{T}'
\end{tikzcd}
$$

The transfer of information between $\mathbb{T}$ and $\mathbb{T}'$ takes place by expressing topos-theoretic *invariants* (that is, properties of (Grothendieck) toposes or constructions on them which are stable under categorical equivalence) defined on their common classifying topos directly in terms of the theories $\mathbb{T}$ and $\mathbb{T}'$. This is usually done by associating with each of the two theories a site of definition for its classifying topos (for example, its geometric syntactic site) and then considering invariants on the common classifying topos from the points of view of the two sites of definition. More precisely, suppose that $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ are two sites of definition for the same topos, and that I is a topos-theoretic invariant. Then one can seek *site characterizations* for I, that is, in the case I is a property (the case of I being a 'construction' admits an analogous treatment), logical equivalences of the kind 'the topos $\mathcal{E}$ satisfies I if and only if $(\mathcal{C}, J)$ satisfies a property $P_{(\mathcal{C},J)}$ (written in the language of the site $(\mathcal{C}, J)$)' and, similarly for $(\mathcal{D}, K)$, logical equivalences of the kind 'the topos $\mathbf{Sh}(\mathcal{D}, K)$ satisfies I if and only if $(\mathcal{D}, K)$ satisfies a property $Q_{(\mathcal{D},K)}$':



Clearly, such characterizations immediately lead to a logical equivalence between the properties $P_{(\mathcal{C},J)}$ and $Q_{(\mathcal{D},K)}$, which can thus be seen as different manifestations of a unique property, namely I, in the context of the two different sites $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$.

In fact, one does not necessarily need 'if and only if' site characterizations in order to build 'bridges': in order to establish an implication between a property $P_{(\mathcal{C},J)}$ of a site $(\mathcal{C}, J)$ and a property $Q_{(\mathcal{D},K)}$ of another site of definition $(\mathcal{D}, K)$ of the same topos, it suffices to find an invariant I such that, on the one hand, $P_{(\mathcal{C},J)}$ implies I on $\mathbf{Sh}(\mathcal{C}, J)$ and, on the other hand, I on $\mathbf{Sh}(\mathcal{D}, K)$ implies $Q_{(\mathcal{D},K)}$.

The 'bridge' technique allows us to interpret and study many correspondences, dualities and equivalences arising in different fields of mathematics by means of the investigation of the way topos-theoretic invariants express themselves in terms of sites. In other words, the representation theory of Grothendieck toposes becomes a sort of 'meta-theory of mathematical duality' which makes it possible to effectively compare distinct mathematical theories with each other and transfer knowledge between them. In the following sections we discuss in more detail the subject of *Morita equivalences*, which play the role of 'decks' of our 'bridges', and of *site characterizations* for topos-theoretic invariants, which constitute their 'arches'.

Incidentally, it should be noted that this method could be generalized to the case of 'bridges' whose decks are provided by some kind of relationship between toposes which is not necessarily an equivalence, in the presence of properties or constructions of toposes which are invariant with respect to such a relation. Nonetheless, the advantage of focusing on Morita equivalences is twofold; on one hand, due to the fact that every property expressed in categorical language is automatically invariant with respect to categorical equivalence, we dispose of an unlimited number of invariants readily available for consideration, while on the other hand Morita equivalences realize a *unification* of 'concrete' properties of different theories by means of their interpretation as different *manifestations* of a unique property lying at the topos-theoretic level.

2.2.2 Decks of 'bridges': *Morita equivalences*

Let us first recall the following classical definition (cf. sections D1.4 and D3.1 of [55]):

Definition 2.2.1 Two geometric theories $\mathbb{T}$ and $\mathbb{T}'$ are said to be *Morita-equivalent* if they have equivalent classifying toposes, i.e. if they have equivalent categories of models in every Grothendieck topos $\mathcal{E}$, naturally in $\mathcal{E}$, that is for each Grothendieck topos $\mathcal{E}$ there is an equivalence of categories

$$\tau_{\mathcal{E}} : \mathbb{T}\text{-mod}(\mathcal{E}) \to \mathbb{T}'\text{-mod}(\mathcal{E})$$

such that for any geometric morphism $f : \mathcal{F} \to \mathcal{E}$ the following diagram commutes (up to isomorphism):

$$\begin{tikzcd}
\mathbb{T}\text{-mod}(\mathcal{E}) \arrow[r, "\tau_{\mathcal{E}}"] \arrow[d, "f^*"'] & \mathbb{T}'\text{-mod}(\mathcal{E}) \arrow[d, "f^*"] \\
\mathbb{T}\text{-mod}(\mathcal{F}) \arrow[r, "\tau_{\mathcal{F}}"] & \mathbb{T}'\text{-mod}(\mathcal{F}).
\end{tikzcd}$$

Note that 'to be Morita-equivalent to each other' defines an equivalence relation on the collection of all geometric theories.

Given the level of technical sophistication of this definition, it is reasonable to wonder whether Morita equivalences naturally arise in mathematics and, if that is the case, whether there are systematic ways of 'generating' them. The following remarks are meant to show that the answer to both questions is positive.

- If two geometric theories $\mathbb{T}$ and $\mathbb{T}'$ have equivalent categories of models in the category **Set** then, provided that the given categorical equivalence is established by only using constructive logic (that is, by avoiding in particular the law of excluded middle and the axiom of choice) and geometric constructions (that is, by only using set-theoretic constructions which involve finite limits and arbitrary colimits, equivalently which admit a syntactic formulation involving only equalities, finite conjunctions, (possibly) infinitary disjunctions and existential quantifications), one can reasonably expect the original equivalence to 'lift' to a Morita equivalence between $\mathbb{T}$ and $\mathbb{T}'$. Indeed, as we have seen in section 1.3.5, a Grothendieck topos behaves logically as a 'generalized universe of sets' in which one can perform most of the classical set-theoretic arguments and constructions, with the significant exception of those requiring non-constructive principles. One could thus expect to be able to generalize the original equivalence between the categories of set-based models of the two theories to the case of models in arbitrary Grothendieck toposes; the fact that the constructions involved in the definition of the equivalence are geometric would then ensure that the above-mentioned naturality condition for Morita equivalences be satisfied (since geometric constructions are preserved by inverse image functors of geometric morphisms). As examples of 'liftings' of naturally arising categorical equivalences to Morita equivalences we mention the Morita equivalence between MV-algebras and abelian ℓ-groups with strong unit (cf. [30]) and that between abelian ℓ-groups and perfect MV-algebras (cf. [31]).

- Two cartesian (in particular, finitary algebraic) theories $\mathbb{T}$ and $\mathbb{T}'$ have equivalent categories of models in **Set** if and only if they are Morita-equivalent (or, equivalently, if and only if their cartesian syntactic categories are equivalent). Indeed, $\mathbb{T}$-mod(**Set**) $\simeq$ $\mathbb{T}'$-mod(**Set**) if and only if f.p.$\mathbb{T}$-mod(**Set**) $\simeq$ f.p.$\mathbb{T}'$-mod(**Set**), if and only if $\mathcal{C}_{\mathbb{T}}^{\text{cart}} \simeq \mathcal{C}_{\mathbb{T}'}^{\text{cart}}$, if and only if $\mathcal{E}_{\mathbb{T}} \simeq \mathcal{E}_{\mathbb{T}'}$ (cf. Theorem 2.1.21).
- If two geometric (resp. regular, coherent) theories have equivalent geometric (resp. regular, coherent) syntactic categories (i.e. they are bi-interpretable in the sense of Definition 2.1.13) then they are Morita-equivalent. This follows at once from Theorem 2.1.8 by observing that the 'logical' topologies in the syntactic sites are defined intrinsically in terms of the categorical structure present on the relevant syntactic categories. Anyway, as can be naturally expected, the most interesting Morita equivalences do not arise from bi-interpretations.

  In particular, if two finitary first-order theories are bi-interpretable (in the sense of classical model theory) then their Morleyizations (in the sense of Proposition 2.1.1) are Morita-equivalent. In fact, from Proposition 2.1.2(ii) it follows at once that the syntactic Boolean pretoposes of two classical first-order theories are equivalent if and only if their Morleyizations are Morita-equivalent.
- Two associative rings with unit are Morita-equivalent (in the classical, ring-theoretic, sense) if and only if the algebraic theories axiomatizing the (left) modules over them are Morita-equivalent (in the topos-theoretic sense). Indeed, by the second remark above, these theories are Morita-equivalent if and only if their categories of set-based models are equivalent, that is if and only if the categories of (left) modules over the two rings are equivalent. Specifically, for each associative ring with unit R, the theory axiomatizing its (left) R-modules can be defined as the theory obtained from the algebraic theory of abelian groups by adding one unary function symbol for each element of the ring and writing down the obvious equational axioms which express the conditions in the definition of an R-module.
- Other notions of Morita equivalence for various kinds of algebraic or geometric structures considered in the literature can be naturally reformulated as equivalences between different representations of the same topos, and hence as Morita equivalences between different geometric theories. For instance:
  - Two topological groups are Morita-equivalent (in the sense of [65]) if and only if their toposes of continuous actions are equivalent. Natural analogues of this notion for topological and localic groupoids have been studied by several authors and a summary of the main results is contained in section C5.3 of [55].
  - Two small categories are Morita-equivalent (in the sense of [43]) if and only if the corresponding presheaf toposes are equivalent, that

is, if and only if their Cauchy completions (also called Karoubian completions) are equivalent (cf. [10]).

– Two inverse semigroups are Morita-equivalent (in the sense of [70] or, equivalently, of [44]) if and only if their classifying toposes (as defined in [45]) are equivalent (cf. [44]).

- Categorical dualities or equivalences between 'concrete' categories can often be seen as arising from the process of 'functorializing' Morita equivalences which express structural relationships between each pair of objects corresponding to each other under the given duality or equivalence (cf. for instance [16], [26] and [22]). In fact, the theory of geometric morphisms of toposes provides various natural ways of 'functorializing' bunches of Morita equivalences.
- Different small sites of definition for a given topos can be interpreted logically as Morita equivalences between different theories (cf. Theorem 2.1.11); in fact, the converse also holds, in the sense that any Morita equivalence between two geometric theories gives rise canonically to two different sites of definition of their common classifying topos. The representation theory of Grothendieck toposes in terms of sites and, more generally, any technique that allows one to obtain different sites of definition or representations for a given topos (for example, Theorem 1.1.8) thus constitutes a tool for generating Morita equivalences.
- The usual notions of *spectra* for mathematical structures can be naturally interpreted in terms of classifying toposes, and the resulting sheaf representations as arising from Morita equivalences between 'algebraic' and 'topological' representations of such toposes. More specifically, Cole's general theory of spectra [34] (cf. also section 6.5 of [50] for a succinct overview of this theory) is based on the construction of classifying toposes of suitable theories. Coste introduced in [38] alternative representations of such classifying toposes, identifying in particular simple sets of conditions under which they can be represented as toposes of sheaves on a topological space; he then derived from the equivalence between two of these representations, one of essentially algebraic nature and the other of topological nature, a criterion for the canonical homomorphism from the given algebraic structure to the structure of global sections of the associated structure sheaf to be an isomorphism.
- The notion of Morita equivalences materializes in many situations the intuitive feeling of 'looking at the same thing in different ways', meaning, for instance, describing the same structure(s) in different languages or constructing a given object in different ways. Concrete examples of this general remark can be found for instance in [16] and [22], where the different constructions of the Zariski spectrum of a ring, of the Gelfand spectrum of a C^*-algebra and of the Stone-Čech compactification of a topological space are interpreted as Morita equivalences between different theories.

- Different ways of looking at a given mathematical theory can often be formalized as Morita equivalences. Indeed, a different way of describing the structures axiomatized by a certain theory often gives rise to a theory written in a different language whose models (in any Grothendieck topos) can be identified, in a natural way, with those of the former theory and which is therefore Morita-equivalent to it.

- A geometric theory *alone* generates an infinite number of Morita equivalences via its 'internal dynamics'. In fact, any way of looking at a geometric theory as an extension of a geometric theory written over its signature provides a different representation of its classifying topos as a subtopos of the classifying topos of the latter theory (cf. the duality theorem of Chapter 3).

- As we remarked in section 1.1.2, different separating sets of objects for a given topos give rise to different sites of definition for it; indeed, for any separating set of objects $\mathcal{C}$ of a Grothendieck topos we have an equivalence $\mathcal{E} \simeq \mathbf{Sh}(\mathcal{C}, J_{\mathcal{E}}|_{\mathcal{C}})$, where $J_{\mathcal{E}}|_{\mathcal{C}}$ is the Grothendieck topology on $\mathcal{C}$ induced by the canonical topology $J_{\mathcal{E}}$ on $\mathcal{E}$. In particular, for any topological space X and any basis $\mathcal{B}$ for it, we have an equivalence $\mathbf{Sh}(X) \simeq \mathbf{Sh}(\mathcal{B}, J_{\mathbf{Sh}(X)}|_{\mathcal{B}})$ (cf. [16] and [22] for examples of dualities arising from Morita equivalences of this form where the induced topologies $J_{\mathbf{Sh}(X)}|_{\mathcal{B}}$ can be characterized intrinsically by means of topos-theoretic invariants).

2.2.3 Arches of 'bridges'*: site characterizations*

As we remarked above, the 'arches' of topos-theoretic 'bridges' should be provided by site characterizations for topos-theoretic invariants, that is results connecting invariant properties of (resp. constructions on) toposes and properties of (resp. constructions on) their sites of definition (written in their respective languages).

It thus becomes crucial to investigate the behaviour of topos-theoretic invariants with respect to sites. As a matter of fact, this behaviour is often very natural, in the sense that topos-theoretic invariants generally admit natural site characterizations. For instance, 'if and only if' characterizations for a wide class of geometric invariants of toposes, notably including the property of a topos being atomic (resp. locally connected, localic, equivalent to a presheaf topos, compact, two-valued), were obtained in [21]. On the other hand, it was shown in [25] that a wide class of logically inspired invariants of toposes, obtained by interpreting first-order formulae written in the language of Heyting algebras in the internal Heyting algebra given by the subobject classifier (cf. Theorem 1.3.29), admit elementary 'if and only if' site characterizations. Moreover, we shall see in Chapters 3 and 4 that several notable invariants of subtoposes admit natural site chacterizations as well as explicit logical descriptions in terms of the theories classified by them. Topos-theoretic invariants relevant in algebraic geometry and homotopy theory, such as cohomology and homotopy groups of toposes, also admit, at least in many important cases, natural characterizations in terms of sites.

It should be noted that, whilst it is often possible to obtain, by using topos-theoretic methods, site characterizations for topos-theoretic invariants holding for large classes of sites, such criteria can be highly non-trivial as far as their mathematical depth is concerned (since the representation theory of toposes is by all means a non-trivial subject) and hence, when combined with specific Morita equivalences to form 'bridges', they can lead to deep results on the relevant theories, especially when the given Morita equivalence is a non-trivial one. These insights can actually be quite surprising when observed from a concrete point of view (that is from the point of view of the two theories related by the Morita equivalence), since a given topos-theoretic invariant may manifest itself in very different ways in the context of different sites. For instance, as shown in [13], for any (small) site $(\mathcal{C}, J)$ the topos $\mathbf{Sh}(\mathcal{C}, J)$ satisfies the invariant property of being De Morgan if and only if for any object c of the category $\mathcal{C}$ and any J-closed sieve R on c, the sieve

$$\{ f : d \to c \mid (f^*(R) = R_d) \text{ or } (\text{for any } g : e \to d, g^*(f^*(R)) = R_e \Rightarrow g \in R_d)\}$$

is J-covering, where $R_c := \{ f : d \to c \mid \emptyset \in J(d)\}$ (for any $c \in \mathcal{C}$).

This property specializes, on a presheaf topos $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$, to the property of $\mathcal{C}$ to satisfy the right Ore condition, and on a topos $\mathbf{Sh}(X)$ of sheaves on a topological space X, to the property of X to be extremely disconnected (i.e. the property that the closure of any open set is open) – cf. also [52]. We shall see in the book many other examples enlightening the natural behaviour of topos-theoretic invariants with respect to sites.

The 'centrality' of topos-theoretic invariants in mathematics is well illustrated by the fact that, in spite of their apparent remoteness from the more 'concrete' objects of study in mathematics, once translated at the level of sites or theories, they often specialize to construction of natural mathematical or logical interest. Besides cohomology and homotopy groups of toposes, whose 'concrete' instantiations in the context of topological spaces and schemes have been of central importance in topology and algebraic geometry throughout the last decades, a great number of other invariants, including many which might seem too abstract to be connected to any problem of natural mathematical interest, can be profitably used to shed light on classical theories. For example, a logically inspired construction such as the *DeMorganization* of a topos (introduced in [13] as the largest dense subtopos of a given topos satisfying De Morgan's law) was shown in [29] to yield, when applied to a specific topos such as the classifying topos of the coherent theory of fields, the classifying topos of a very natural mathematical theory, namely the theory of fields of finite characteristic which are algebraic over their prime field. The author's papers contain several other examples, some of which will be presented in the later chapters of the book.

It should be noted that the arches of our 'bridges' need not necessarily be 'symmetric', that is arising from an instantiation in the context of two given sites of a unique site characterization holding for both of them. As an example, take the property of a topos to be coherent: this invariant admits an 'elementary' site characterization holding for all trivial sites (i.e. sites whose underlying Grothendieck topology is trivial), and the following implication holds: if a theory is coherent then its classifying topos is coherent. These criteria can for instance be combined to obtain a 'bridge' yielding a result on coherent theories classified by a presheaf topos.

The level of mathematical 'depth' of the results obtained by applying the 'bridge' technique can vary enormously, depending on the complexity of the site characterizations and of the given Morita equivalence. Still, as we shall see at various points of the book, even very simple invariants applied to easily established Morita equivalences can yield surprising insights which would be hardly attainable, or not even imaginable, otherwise.

Lastly, it is worth noting that one can apply the 'bridge' technique, described in section 2.2.1 for sites, also in the case of alternative mathematical objects used for representing Grothendieck toposes (such as topological or localic groupoids, or quantales), replacing site characterizations with appropriate characterizations of the given invariant in terms of such objects.

2.2.4 *Some simple examples*

In this section we shall derive, as applications of the 'bridge' technique, some logical results which settle questions left unanswered in the previous sections of the chapter (cf. Remarks 1.2.14 and 2.1.18(a)).

Recall that for any fragment of geometric logic there is a corresponding notion of provability for theories in that fragment; for instance, we have a notion of provability of regular sequents in regular logic and a notion of provability of coherent sequents in coherent logic. A natural question is whether these notions of provability are compatible with each other, that is, if the notion of provability in a given fragment of logic specializes to the notion of provability in a smaller fragment when applied to sequents lying in that fragment (note that a theory in a given fragment can always be considered as a theory in any larger fragment). Similar questions can be posed concerning the notion of bi-interpretability and of completeness of a theory in a given fragment of logic with respect to its set-based models.

There is a natural topos-theoretic way of dealing with all these questions, which exploits the fact that for a theory in a given fragment of geometric logic one has multiple syntactic representations of its classifying topos, each corresponding to a particular larger fragment of logic in which the theory can be considered (cf. Theorem 2.1.8). For example, given a coherent theory $\mathbb{T}$ over a signature Σ, we have two syntactic representations of its classifying topos: as the category of sheaves $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ on the coherent syntactic site $(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ of $\mathbb{T}$ and as the category of sheaves $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ on the geometric syntactic site $(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ of $\mathbb{T}$. Transferring appropriate invariants across such different representations will yield the desired results:

$$
\begin{tikzcd}[column sep=small]
 & \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \arrow[dl, dash, dashed] \arrow[dr, dash, dashed] & \\
(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}}) & & (\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})
\end{tikzcd}
.
$$

We shall address these questions one by one, highlighting in each case the relevant topos-theoretic invariants and site characterizations.

Below, by a *fragment* L of geometric logic we mean either cartesian, regular, coherent or geometric logic. Clearly, each of this fragments is contained in the successive in the list. By a sequent in cartesian (resp. regular, coherent, geometric) logic we mean a cartesian (resp. regular, coherent, geometric) sequent.

Theorem 2.2.2 *Let $\mathbb{T}$ be a theory in a fragment L of geometric logic and σ a sequent in L over its signature. Then σ is provable in $\mathbb{T}$, regarded as a theory in L, if and only if it is provable in $\mathbb{T}$, regarded as a theory in any fragment of geometric logic containing L.*

Proof We shall give the proof of the theorem in the case when $\mathbb{T}$ is a coherent theory, regarded in both coherent logic and geometric logic. The other cases can be proved in a completely analogous way.

$$
\begin{tikzcd}[column sep=small, row sep=small]
 & \textit{Validity in the} & \\
 & \textit{universal model of } \mathbb{T} & \\
 & \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \arrow[dl, dash, dashed] \arrow[dr, dash, dashed] & \\
(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}}) & & (\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \\
\textit{Provability in } \mathbb{T} & & \textit{Provability in } \mathbb{T} \\
\textit{w.r.t. coherent logic} & & \textit{w.r.t. geometric logic}
\end{tikzcd}
$$

From the fact that the Yoneda embedding $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ is conservative, it follows that a coherent sequent over Σ is valid in the universal model of $\mathbb{T}$ lying in the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ if and only if it is valid in the universal model $M_{\mathbb{T}}^{\mathrm{coh}}$ of $\mathbb{T}$ lying in its coherent syntactic category $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}$, i.e. if and only if it is provable in $\mathbb{T}$ by using coherent logic (cf. Theorem 1.4.5 and Remark 2.1.9). Similarly, one obtains that a geometric sequent over Σ is valid in the universal model of $\mathbb{T}$ lying in the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ if and only if it is provable in $\mathbb{T}$ by using geometric logic. But the property of validity of a given sequent in the universal model of a geometric theory is a topos-theoretic invariant, whence a coherent sequent is provable in $\mathbb{T}$ by using coherent logic if and only if it is provable in $\mathbb{T}$ by using geometric logic, as required. $\square$

Let us now turn to the topic of 'classical completeness' of theories in fragments of geometric logic. In section 1.4.5 we introduced the property, for a theory considered within a given fragment of geometric logic, of having enough set-based models. It is natural to wonder whether this property depends on the fragment of logic in which the theory is considered. We shall now see, again by applying the 'bridge technique', that this is not the case.

Theorem 2.2.3 *Let $\mathbb{T}$ be a theory in a given fragment L of geometric logic. Then $\mathbb{T}$ has enough set-based models, regarded as a theory in L, if and only if it has enough set-based models, regarded as a theory in any larger fragment of geometric logic containing L.*

Proof As we did for the previous result, we shall give the proof of the theorem in the case of a coherent theory $\mathbb{T}$, regarded both in coherent logic and in geometric logic, the other cases being completely analogous.

The result naturally arises from the transfer of an invariant, namely the property of a topos of having enough points, across the two different representations $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ of the classifying topos of $\mathbb{T}$. Indeed, in terms of the site $(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ the invariant rephrases as the condition that every coherent sequent over Σ which is satisfied in all the set-based models of $\mathbb{T}$ should be provable in $\mathbb{T}$ by using coherent logic, while in terms of the site $(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ it rephrases as the condition that every geometric sequent over Σ which is satisfied in all the set-based models of $\mathbb{T}$ should be provable in $\mathbb{T}$ by using geometric logic:

Property of having enough points

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

$(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$
Classical completeness w.r.t. coherent logic

$(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$
Classical completeness w.r.t. geometric logic

The above-mentioned site characterizations for the property of the classifying topos of having enough points can be proved as follows. By definition, $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ has enough points if and only if the inverse image functors f^* of the geometric morphisms $f : \mathbf{Set} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ are jointly conservative. Now, since the geometric morphism $f_M : \mathbf{Set} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ corresponding to a $\mathbb{T}$-model M in **Set** satisfies $f^*(y^{\mathrm{coh}}(M_{\mathbb{T}}^{\mathrm{coh}})) = M$ (where $M_{\mathbb{T}}^{\mathrm{coh}}$ is the universal model of $\mathbb{T}$ lying in the coherent syntactic category $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}$ of $\mathbb{T}$ – as in Definition 1.4.4 and Remark 1.4.6 – and $y^{\mathrm{coh}} : \mathcal{C}_{\mathbb{T}}^{\mathrm{coh}} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ is the Yoneda embedding) we have that, by Theorem 1.4.5 and the fact that y^{coh} is conservative, if the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ has enough points then any coherent sequent over Σ which is satisfied in every $\mathbb{T}$-model M in **Set** is satisfied in $M_{\mathbb{T}}^{\mathrm{coh}}$ i.e. provable in $\mathbb{T}$ within coherent logic. Conversely, the fact that $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}$ is a separating set of objects for $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ easily implies that if $\mathbb{T}$ satisfies classical completeness with respect to coherent logic then $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}, J_{\mathbb{T}}^{\mathrm{coh}})$ has enough points. A similar characterization can be obtained for the geometric syntactic site of $\mathbb{T}$ (cf. Proposition 4.2.20 below). $\square$

Finally, let us discuss the independence of the notion of bi-interpretability of theories (in the sense of Definition 1.4.12) from the fragment of logic in which they are considered.

Theorem 2.2.4 *Let $\mathbb{T}$ and $\mathbb{S}$ be two theories in a fragment L of geometric logic. Then $\mathbb{T}$ and $\mathbb{S}$ are bi-interpretable within the fragment L if and only if they are bi-interpretable within any fragment of geometric logic containing L.*

Proof We shall first give the proof of the theorem in the case of two regular theories $\mathbb{T}$ and $\mathbb{S}$, considered both in regular logic and in coherent logic, and then discuss how to modify the proof in the other cases.

The 'only if' direction follows immediately from the universal properties of regular and coherent syntactic categories. Indeed, by Theorem 1.4.10 any equivalence $\mathcal{C}_{\mathbb{T}}^{\text{reg}} \simeq \mathcal{C}_{\mathbb{S}}^{\text{reg}}$ induces an equivalence between the categories of models of $\mathbb{T}$ and $\mathbb{S}$ in any regular, and hence in particular in any coherent, category $\mathcal{D}$, naturally in $\mathcal{D}$. The 2-dimensional Yoneda lemma (cf. section 3.1.1) thus yields an equivalence $\mathcal{C}_{\mathbb{T}}^{\text{coh}} \simeq \mathcal{C}_{\mathbb{S}}^{\text{coh}}$, as desired.

The 'if' direction can be proved by using the 'bridge' technique, as follows.

In order to deduce that $\mathcal{C}_{\mathbb{T}}^{\text{reg}} \simeq \mathcal{C}_{\mathbb{S}}^{\text{reg}}$ from $\mathcal{C}_{\mathbb{T}}^{\text{coh}} \simeq \mathcal{C}_{\mathbb{S}}^{\text{coh}}$, it will be sufficient to characterize $\mathcal{C}_{\mathbb{T}}^{\text{reg}}$ as a full subcategory of $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$ in an invariant way. Since both these categories fully embed in the classifying topos of $\mathbb{T}$ (as the topologies $J_{\mathbb{T}}^{\text{reg}}$ and $J_{\mathbb{T}}^{\text{coh}}$ are subcanonical) it would be enough to characterize in invariant terms the objects of the classifying topos of $\mathbb{T}$ which come from the regular syntactic site of $\mathbb{T}$ among those which come from the coherent syntactic site of $\mathbb{T}$. We notice that all the objects of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathbb{T}}^{\text{reg}})$ coming from $\mathcal{C}_{\mathbb{T}}^{\text{reg}}$ are supercompact (cf. Definition 2.1.14). Indeed, the property of the site $(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathbb{T}}^{\text{reg}})$ to be regular implies that the image under the Yoneda embedding $y^{\text{reg}} : \mathcal{C}_{\mathbb{T}}^{\text{reg}} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathbb{T}}^{\text{reg}})$ of any object of $\mathcal{C}_{\mathbb{T}}^{\text{reg}}$ is a supercoherent and *a fortiori* supercompact object of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathbb{T}}^{\text{reg}})$ (cf. Proposition 10.8.1(ii)).

On the other hand, one can easily characterize in terms of the coherent syntactic site of $\mathbb{T}$ the property of an object $\{\vec{x} \,.\, \phi\}$ of $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$ to be sent by the Yoneda embedding $y^{\text{coh}} : \mathcal{C}_{\mathbb{T}}^{\text{coh}} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, J_{\mathbb{T}}^{\text{coh}})$ to a supercompact object of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, J_{\mathbb{T}}^{\text{coh}})$: $y^{\text{coh}}(\{\vec{x} \,.\, \phi\})$ is supercompact in $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, J_{\mathbb{T}}^{\text{coh}})$ if and only if for any finite family $\{\psi_i(\vec{x}) \mid i \in I\}$ of coherent formulae in the same context, $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} \psi_i)$ provable in $\mathbb{T}$ implies $(\phi \vdash_{\vec{x}} \psi_i)$ provable in $\mathbb{T}$ for some $i \in I$. Let us refer to this property of objects $\{\vec{x} \,.\, \phi\}$ of $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$ as to 'finite regularity'. So, we have that an object of $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$ is finitely regular if and only if every finite covering of it by subobjects in $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$ is trivial.

$$
\begin{tikzcd}
 & \underset{\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathbb{T}}^{\text{reg}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, J_{\mathbb{T}}^{\text{coh}})}{\overset{\textit{Supercompact object}}{}} \arrow[dr, dashed] & \\
\underset{\textit{Every object}}{(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathbb{T}}^{\text{reg}})} \arrow[ur, dashed] & & \underset{\textit{Finitely regular object}}{(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, J_{\mathcal{C}_{\mathbb{T}}}^{\text{coh}})}
\end{tikzcd}
$$

Now, it can easily be proved by induction on the structure of coherent formulae that every coherent formula is provably equivalent in coherent logic to a finite disjunction of regular formulae in the same context. This clearly implies that every finitely regular coherent formula $\{\vec{x}\,.\,\phi\}$ is isomorphic in $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}$ to a regular formula. We can thus conclude that the full subcategory of $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}}$ on the finitely regular objects is equivalent to the category $\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}}$, and similarly for the theory $\mathbb{S}$. It follows that the equivalence $\mathcal{C}_{\mathbb{T}}^{\mathrm{coh}} \simeq \mathcal{C}_{\mathbb{S}}^{\mathrm{coh}}$ restricts to an equivalence $\mathcal{C}_{\mathbb{T}}^{\mathrm{reg}} \simeq \mathcal{C}_{\mathbb{S}}^{\mathrm{reg}}$, as desired.

In the case of a cartesian theory considered in a larger fragment of geometric logic, the notion to use in the place of that of supercompact object is that of irreducible object (the only difference in the proof being that, in order to show that every formula which is sent by the Yoneda embedding to an irreducible object of the classifying topos is isomorphic in the relevant syntactic category to a cartesian formula, one uses Proposition 2.1.15).

In the case of coherent theories, the notion to use is instead that of compact object. □

2.2.5 *A theory of 'structural translations'*

Let us conclude this chapter with a few methodological remarks.

The view underlying the 'bridge' technique consists in regarding a topos as an object which, together with all its different representations, embodies a great amount of relationships existing between the different theories classified by it. Any topos-theoretic invariant behaves in this context like a 'pair of glasses' which allows us to discern information which is 'hidden' in a given Morita equivalence. Toposes can thus act as 'universal translators' across different mathematical theories which have the same 'semantic content'.

From a technical point of view, the main reason for the effectiveness of the 'bridge' technique is twofold: on the one hand, as we have argued in section 2.2.3, topos-theoretic invariants usually manifest themselves in significantly different ways in the context of different sites; on the other hand, due to the very well-behaved nature of the representation theory of Grothendieck toposes, site characterizations are often derivable by means of rather canonical 'calculations'.

Unlike the traditional, 'dictionary-oriented' method of translation based on a 'renaming', according to a given 'dictionary', of the primitive constituents of the information as expressed in a given language, the 'invariant-oriented' translations realized by topos-theoretic 'bridges' consist of 'structural unravelings' of invariants across different representations of the relevant toposes. Interestingly, for the transfer of 'global' properties of toposes, it is only the *existence* of a Morita equivalence that matters, rather than its explicit description, since, by its very definition, a topos-theoretic invariant is stable under *any* categorical equivalence. If one wants to establish more 'specific' results, one can use invariant properties of *objects* of toposes rather than properties of the whole toposes, in which case the knowledge of at least the 'local' behaviour of the Morita equivalence at such objects is needed, but for investigating most of the 'global' properties of theories such information is not at all necessary.

We have already hinted above to the fact that there is a strong element of *automatism* implicit in the 'bridge' technique. In fact, in order to obtain insights on the Morita equivalence under consideration, in many cases one can just readily apply to it general characterizations connecting properties of sites and topos-theoretic invariants. Still, the results generated in this way are in general non-trivial; in some cases they can be rather 'weird' according to the usual mathematical standards (although they might still be quite deep) but, with a careful choice of Morita equivalences and invariants, one can easily get interesting and natural mathematical results. In fact, a lot of information that is not visible with the usual 'glasses' is revealed by the application of this machinery.

The range of applicability of the 'bridge' technique is very broad within mathematics, by the very generality of the notion of topos (and of that of geometric theory). Through this method, results are generated transversally to the various mathematical fields in a 'uniform' way which is determined by the form of the toposes involved and the invariants considered on them. Notice that this way of doing mathematics is inherently 'upside down': instead of starting with simple ingredients and combining them to build more complicated structures, one assumes as primitive ingredients rich and sophisticated (meta-)mathematical entities, namely Morita equivalences and topos-theoretic invariants, and proceeds to extract from them 'concrete' information relevant for classical mathematics.